Economics 880 – PSet 3

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1 Model Comparison

Table 1: Stationary Equilibrium: Model Comparison

Variable	Standard	TV1 ($\alpha = 1$)	TV1 ($\alpha = 2$)
Price Level	0.7390	0.6908	0.7195
Mass of Incumbents	8.3207	9.5489	8.3505
Mass of Entrants	2.6386	4.2232	3.5112
Mass of Exits	2.6386	4.2232	3.5112
Aggregate Labor	179.8332	188.8909	182.6240
Labor of Incumbents	142.6260	139.5075	136.6501
Labor of Entrants	37.2072	49.3834	45.9739
Fraction of Labor in Entrants	0.2069	0.2614	0.2517

The TV1 specifications generate lower equilibrium prices, greater firm turnover, and higher aggregate employment relative to the standard model. These differences can be understood through three mechanisms.

First, introducing action-specific shocks increases the ex-ante value of operating a firm, since firms choose whether to exit or continue after observing these shocks. This makes entry more attractive at a given output price. Consequently, the equilibrium price must fall to restore the free-entry condition. This effect is stronger when α is smaller, corresponding to greater variance in the action-specific shocks.

Second, the TV1 shocks weaken the relationship between productivity and exit. In the standard model, all firms in the lowest productivity state exit, while firms in higher productivity states survive. Under TV1, some low-productivity firms survive while some higher-productivity firms exit. In our calibration, this increases firm turnover and reduces average firm lifetimes. Since entry must equal exit in a stationary equilibrium, the higher rate of exit is accompanied by greater entry. As a result, the fraction of aggregate employment accounted for by entrants increases.

Finally, aggregate employment increases despite the lower equilibrium price, which reduces labor demand at the individual-firm level. This reflects adjustments along both the extensive margin of firm entry and the household labor supply margin. In particular, aggregate operating profits decline in the TV1 specifications. Since household labor supply satisfies

$$N^s = \frac{1}{A} - \Pi,$$

lower profit income induces households to supply more labor, supporting higher equilibrium employment.

As α increases, the variance of the action-specific shocks decreases, and the TV1 model approaches the deterministic specification. Accordingly, the equilibrium price, aggregate employment, and firm turnover generally move toward their standard-model values. However, individual equilibrium moments need not change monotonically because prices, entry, and the stationary distribution adjust jointly.

2 Decision Rules of Exit

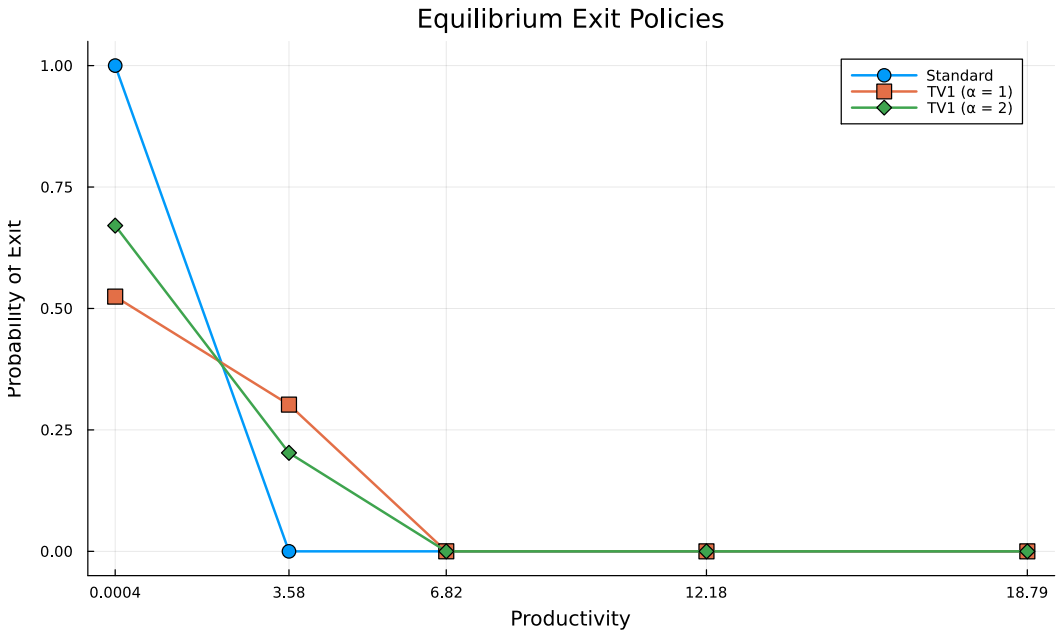


Figure 1: Equilibrium Exit Policies Across Model Specifications

In the standard model, exit decisions are deterministic: all firms in the lowest productivity state exit, while firms in higher productivity states remain. This follows because firms with identical productivity have identical continuation values and therefore make the same exit decision.

Introducing TV1 shocks makes exit decisions probabilistic, allowing firms with identical productivity to make different choices. As a result, some low-productivity firms survive while some higher-productivity firms exit. The exit probabilities remain decreasing in productivity, since more productive firms have higher expected continuation values. However, the relationship becomes smoother rather than following a strict cutoff rule.

The parameter α governs the variance of the action-specific shocks. As α increases, the variance decreases, and firms are more likely to choose the action with the highest fundamental value. Consequently, the TV1 exit probabilities approach the deterministic decision rule as α increases.

3 Fixed Operating Cost Comparative Statics

Increasing the fixed operating cost from $c_f = 10$ to $c_f = 15$ generally increases the probability of exit, particularly among low-productivity firms. Higher fixed costs reduce operating profits and expected continuation values, making continued operation less attractive. Although the equilibrium output price adjusts through the free-entry condition, the overall effect in our simulations is an increase in exit.

In the standard model, the higher fixed cost changes the exit threshold: firms in the two lowest productivity states now exit, whereas previously only firms in the lowest state exited. Thus, the increase in fixed

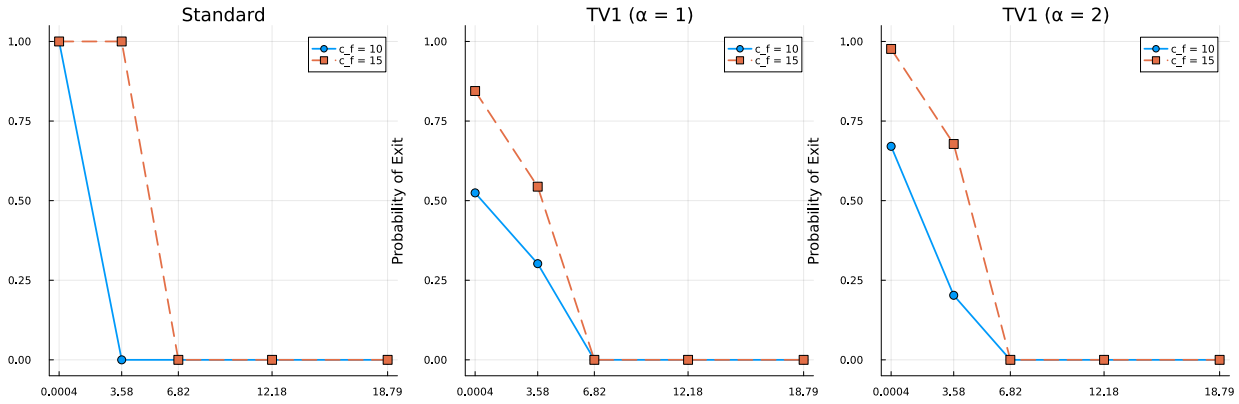


Figure 2: Effect of Increasing Fixed Operating Costs from $c_f = 10$ to $c_f = 15$

costs induces exit among firms that previously found continued operation profitable.

In the TV1 specifications, the adjustment occurs through changes in exit probabilities rather than a discrete change in the exit threshold. For $\alpha = 1$, the exit probabilities in the two lowest productivity states increase from approximately $(0.52, 0.30)$ to $(0.84, 0.55)$. For $\alpha = 2$, they increase from approximately $(0.67, 0.20)$ to $(0.97, 0.67)$. Exit probabilities for firms in the three highest productivity states remain close to zero.

These results illustrate that higher operating costs disproportionately affect firms with low productivity and low continuation values. Moreover, the TV1 model generates a smoother response to changes in operating costs, whereas the standard model exhibits discrete changes in the set of firms choosing to exit.